

H-01-CX-Lagrangian-circulation-scale-time-kill-test v1: Does Lagrangian deformation / circulation force new incompatibility? Single kill audit

ZFL Lab

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MISSION FROZEN: Search only $T^* < \infty \Rightarrow C_H$ with C_H involving material quantities (flow X , gradient $F = \nabla_a X$, circulation Kelvin with viscous term, vortex amplification, critical material time-scale) such that $C_E^{weak} \wedge C_{L^3} \wedge C_H \Rightarrow \perp$ where $C_E^{weak} : \limsup \int_0^t W = +\infty \Rightarrow \int_0^{T^*} W_+ = \infty$, $C_{L^3} : \limsup \|u\|_3 = \infty$.

TREE STATUS: E =CLOSED, F =CLOSED/BLOCKED (F-01 PROVEN NECESSARY NO CONTRADICTION pending quantifiers, F-02 KILLED), G =LOCKED, H =ACTIVE, $C_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ until C_F^{conc} promotion audited.

CANDIDATE C_H DEFINITION ATTEMPT:

Material flow: $\frac{D}{Dt}X(a, t) = u(X(a, t), t)$, $X(a, 0) = a$, $F(a, t) = \nabla_a X(a, t)$, $\partial_t F = (\nabla u)(X(a, t), t)F$, $F(a, 0) = I$.

Vorticity material: $\frac{D\omega}{Dt} = S\omega + \nu\Delta\omega$, $S = \frac{1}{2}(\nabla u + \nabla u^T)$.

Circulation with viscosity: For material loop $C(t) = X(C_0, t)$,

$$\boxed{\frac{D}{Dt}\Gamma_{C(t)} = \nu \oint_{C(t)} \Delta u \cdot dl + \oint_{C(t)} f \cdot dl, \quad \Gamma = \oint_{C(t)} u \cdot dl}$$

Not conservation exact. Inviscid Kelvin $\frac{D\Gamma}{Dt} = 0$ is NOT valid for NS; must keep viscous term. Also Γ not sign-definite.

Deformation bound: $\|F(a, t)\| \leq \exp(\int_0^t \|\nabla u(\cdot, s)\|_\infty ds)$. BKM gives $\int_0^{T^*} \|\omega\|_\infty = \infty$ necessary, $\|\nabla u\|_\infty \geq c\|\omega\|_\infty$. So $\sup_a \|F(a, t)\| \rightarrow \infty$ may follow if blow-up occurs near trajectory a (FTLE $\sup_a \frac{1}{t} \log \|F\| \rightarrow \infty$). However precise quantifier needed: $\forall? \exists a?$ Which norm? Not assumed.

ATTEMPT TO GET $\perp$:

Try C_H : $\exists a, \limsup_{t \uparrow T^*} \|F(a, t)\| = +\infty$ or $\sup_a \|F\| = \infty$ or $\sup_a \text{FTLE} = \infty$.

Does C_H follow necessarily from $T^* < \infty$? Known results: blow-up $\Rightarrow \sup_a \|F(a, t)\| \rightarrow \infty$ and $\sup_a \text{FTLE} \rightarrow \infty$ (Lagrangian obstruction). This uses BKM + $\partial_t F = (\nabla u)F$ and that vorticity max attained near Lagrangian particle. So C_H is necessary, strictly stronger than C_{L^3} ? It involves $\nabla u L^\infty$ not just L^3 . It is at least as strong as BKM.

Now compatibility with $\int_0^{T^*} W_+ = \infty$?

$W_+(t) = \int (\omega^T S \omega)_+$, S part of ∇u . If $\|F\| \rightarrow \infty$, then $\int_0^t \|S\|_\infty \geq c \log \|F\|$ diverges. Since $|W| \leq \|S\|_\infty \|\omega\|_2^2$ and $\|\omega\|_2^2$ may blow up, W_+ can diverge arbitrarily fast while C_H holds. No upper bound.

Quantitative attempt: $W_+ \leq f(\|F\|)$? No, $\|F\|$ lower bound ≥ 1 , W_+ needs upper bound, lower bound cannot imply finite integral. Conversely C_H is lower bound $\|F\| \geq \exp(\dots)$, does not control W_+ from above.

Try circulation: $\Gamma_{C(t)}$ evolution contains $\nu \oint \Delta u$. Does Γ bound W_+ ? Kelvin theorem with viscosity: $|\frac{D\Gamma}{Dt}| \leq \nu \|\nabla \omega\|_{L^1(C)}$? Not controlling global W_+ .

Scale-time: F time-scale $\sim (\|\nabla u\|_\infty)^{-1}$, critical material time-scale goes to 0 if $\|\nabla u\|_\infty \rightarrow \infty$. Does that force $\int W_+ < \infty$? No, W_+ can still integrate to ∞ because $\|\omega\|_2^2$ can grow.

Thus C_H compatible with $C_E^{weak} \wedge C_{L^3}$ and $\int W_+ = +\infty$.

CHECK KILL CONDITIONS (frozen):

- If C_H is just $\frac{D\omega}{Dt} = S\omega + \nu \Delta \omega$ or reformulation of stretching: KILL. We have that, but C_H as $\|F\| \rightarrow \infty$ is consequence of BKM, not just rewriting stretching? It is essentially rewriting BKM in Lagrangian terms: $\int \|\nabla u\|_\infty = \infty \Rightarrow \sup \|F\| = \infty$. This is equivalent to BKM, not new beyond C_E^{weak} ? BKM already implies $\int \|\omega\|_\infty = \infty$, stronger than $\int W_+ = +\infty$? But C_H does not give new upper bound.

- If $C_H \equiv C_{L^3}$ or simply equivalent: Not equivalent, but does not add incompatibility. KILL per second condition if compatible with $\int W_+ = +\infty$.

- If assumes $T^* < \infty \Rightarrow \|\nabla X\| \rightarrow \infty$ without quantifier: we avoid, we specify $\sup_a \|F(a, t)\| \rightarrow \infty$ exists a near blow-up, with precise norm operator norm, trajectory a such that $X(a, t)$ approaches singular point.

- No mixing inviscid Kelvin exact: we keep ν term explicitly, so circulation not conserved, no false conservation.

Therefore:

$$H-01 \text{ yields necessary } C_H : \sup_a \|F(a, t)\| \rightarrow \infty, \sup_a \text{FTLE} \rightarrow \infty, \frac{D\Gamma}{Dt} = \nu \oint \Delta u$$

but

$$C_E^{weak} \wedge C_{L^3} \wedge C_H \text{ compatible, no } \perp$$

$$\text{No quantitative } W_+ \leq f(C_H)$$

C_H is lower bound, W_+ upper needs upper bound.

VERDICT:

$$H-01 = KILLED \text{ as bridge to } \perp$$

but

$$H-01 = OPEN \text{ as PROVEN NECESSARY Lagrangian obstruction, NO CONTRADICTION similar to F-01.}$$

Thus H family does not close main question, no H-02 rescue.

If we want promotion:

$$C_{PROVEN} \text{ could include } C_H^{Lag} : \sup_a \|F\| \rightarrow \infty \text{ as PROVEN NECESSARY if quantifiers proved}$$

$$\text{but central } C_E^{weak} \wedge C_{L^3} \Rightarrow \perp OPEN \text{ remains.}$$

Next family: new axis beyond E,F,H,G.